

AI Contract Auditor - Technical Exercise Write-up

Repository: <https://github.com/HindAljuaid/AI-Contract-Auditor>

Live demo: <https://ai-contract-auditor.streamlit.app>

Approach and how I measured results

I built a hybrid audit pipeline that separates language interpretation from financial decision-making. OpenAI is used to convert heterogeneous contract text into a typed contract-rule schema and, when enabled, to help resolve only unclear service-description mappings. All monetary calculations and final findings are performed deterministically in Python using integer cents, explicit rate periods, and rule-specific checks. The engine covers structural errors, unit/rate mismatches, date rules, duplicates, premiums, discounts, bundles, daily caps, exclusions, and invoice totals.

Hospital 1 was the only labeled dataset, so I used it strictly as a development/calibration benchmark rather than as evidence of out-of-sample performance. I evaluated all 18 error categories with precision, recall, and F1. The final H1 validator achieved 18/18 perfect categories, with 105 true positives, 0 false positives, and 0 false negatives (mean precision = 1.000, recall = 1.000, F1 = 1.000). The repository preserves the notebook, standalone validator, category evaluation, predictions, and line mappings for reproducibility.

Hospitals 2-5 are unlabeled, so I did not report accuracy for them. Instead, I monitored internal audit signals such as mapping coverage, final confidence, flagged counts, and the manual-review queue. These are system outputs, not verified ground truth.

Hospital	Predictions	Flagged	Mapping coverage	Avg. confidence
H2	1,125	141	96.0%	73.6%
H3	932	92	97.6%	72.2%
H4	835	295	92.7%	73.6%
H5	1,050	157	96.8%	77.3%

Where I was uncertain and why

The largest source of uncertainty was contract interpretation rather than arithmetic. Real invoice descriptions do not match contract wording exactly, so unresolved or low-confidence mappings are left visible instead of being forced into a service. Contract clauses that cannot be represented safely in the generic schema are retained as unsupported rules or ambiguities and reduce confidence. Examples include the aggregation scope of cumulative volume thresholds, duplicate-billing clauses, the scope of daily caps, and endpoint interpretation for exclusion windows.

I also treated duplicate invoice IDs conservatively. Multiple physical source records can share one invoice ID, so the pipeline preserves source-row context, links line items to the correct source record where possible, and uses the latest physical record as the canonical submission record. Duplicate records are shown separately and are never summed together. Any unresolved duplicate context remains a review case.

The scored hospitals all contain at least one contract-level unsupported rule, so the application intentionally marks their outputs for human review even when invoice-level mapping coverage is high. I treat the review queue as an uncertainty signal, not an accuracy score. The submission therefore contains my best deterministic predictions while the repository also preserves the contract-rule JSON, review queue, service mappings, and line-level evidence used to generate them.

What I would do differently with another week

- Implement contract-specific adapters for every currently unsupported clause, especially cumulative-volume scope, duplicate-service prohibitions, daily-cap aggregation, and exclusion-window boundary semantics.
- Add clause-level provenance to every extracted rule and require reviewer approval/versioning before an audit is treated as final.
- Create regression tests for every special contract clause and add end-to-end tests that compare exported submissions with expected deterministic evidence.
- Calibrate confidence more rigorously. I would use H1 only for development calibration, then add synthetic perturbation tests or a held-out validation protocol rather than treating H1 perfection as generalization.
- Improve service matching with stronger contract-aware candidate generation and a small reviewed mapping dictionary, while preserving the rule that uncertain mappings remain unresolved rather than guessed.
- Perform a structured manual review of the highest-impact H2-H5 findings, especially large expected-vs-billed differences and low-confidence mappings, before final submission.

Submission artifacts

The final submission.csv follows the repository template exactly and combines Hospitals 2-5 into one file with 3,942 unique invoice rows. The GitHub repository contains the application, reproducible H1 validation evidence, per-hospital contract rules and audit outputs, tests, and the deployed Streamlit demo.